

EV

Elena Vasquez

SENIOR ML ENGINEER · LLMS & GENAI

→ elena.v@mleng.io

→ +1 (512) 834-0571

→ Austin, TX · Remote

→ github.com/elena-vasquez-ml

→ scholar.google/elena-vasquez

ML EXPERTISE

LLMs / Transformers
Expert · 6 years

Generative AI
Expert · 4 years

MLOps / ML Systems
Expert · 5 years

Computer Vision
Advanced · 4 years

RL / RLHF
Advanced · 3 years

STACK

Python

PyTorch

JAX

TF

HuggingFace

vLLM

CUDA

Triton

Ray

Spark

MLflow

W&B

K8s

Docker

CERTIFICATIONS

AWS ML Specialty

GCP Professional ML Eng

NVIDIA Deep Learning

DeepLearning.AI Specialization

EDUCATION

Ph.D. Computer Science

UT Austin
2013 · NLP Lab · NSF Fellow

M.S. Machine Learning

Georgia Tech
2009

B.S. Mathematics + CS

University of Florida
2007 · Summa Cum Laude

LANGUAGES

English

Native

Spanish

Native

Portuguese

Fluent

11+

YRS ML ENG

24

PAPERS PUB.

7.2K

GH STARS

4

PATENTS

Senior ML Engineer specializing in LLMs, generative AI, and production ML systems at scale. 11+ years training

and deploying models across NLP, vision, and multimodal domains. Deep expertise in RLHF, model optimization, and inference efficiency. Led foundational model teams at OpenAI, Meta AI, and Nvidia. Author of 24 papers with 6,000+ citations.

Experience

Staff ML Engineer - LLM Alignment

Mar 2022 – Present

OpenAI · San Francisco, CA

- \$ Core RLHF engineer on GPT-4 and GPT-4o; designed reward model training pipeline processing 2M+ human preference pairs
- \$ Built data quality pipeline for constitutional AI training reducing hallucination rate by 34% on TruthfulQA benchmark
- \$ Developed speculative decoding system achieving 2.8x inference speedup while maintaining output quality parity
- \$ Led 0→1 development of automated red-teaming framework; discovered and resolved 89 critical safety issues pre-launch
- \$ Optimized KV-cache architecture reducing memory consumption by 42% for 100K+ context windows

Research Engineer, Fundamental AI Research

Sep 2018 – Feb 2022

Meta AI (FAIR) · Menlo Park, CA

- \$ Co-developed OPT-175B open-source LLM; contributed model parallelism and activation checkpointing improvements enabling 15% faster training
- \$ Published "Scaling Laws for Neural Language Model Fine-tuning" (ICML 2021) with 1,400+ citations
- \$ Built multimodal image+text model achieving SOTA on VQAv2 and GQA benchmarks; deployed to Instagram for content understanding
- \$ Optimized FAISS vector search library; contributed PR reducing memory usage by 30%, merged into main branch

Senior Deep Learning Engineer

Jul 2013 – Aug 2018

NVIDIA Research · Santa Clara, CA

- \$ Developed CUDA kernels for early Transformer architectures, achieving 3.2x training speedup over naive PyTorch
- \$ Co-designed TensorRT optimizations for BERT inference; reduced latency from 80ms to 8ms (10x)
- \$ Led team building cuDNN convolution algorithms; authored 2 patents on memory layout optimization

Select Publications

Vasquez et al. "Scaling Laws for RLHF in Large Language Models" — NeurIPS 2023

Vasquez, Chen, Li "Efficient Fine-tuning at Scale" — ICML 2021 · 1,400 citations

Vasquez et al. "Mixture of Experts for Multilingual NLP" — ACL 2020 · 890 citations